# Ian Yang

+1 341-766-8282 | ianyang@berkeley.edu | linkedin.com/in/ian--yang

## EDUCATION

**University of California, Berkeley** — **Expected Graduation: 05/2028**

*Bachelor of Science in Electrical Engineering and Computer Sciences (EECS)* — *GPA: 3.63/4.0*

- Relevant Coursework: Deep Learning for Visual Data, Intro to Machine Learning, Artificial Intelligence, Intro to Robotics, LLM Agents, Optimization Models in Engineering, Machine Structures, Data Structures, Efficient Algorithms & Intractable Problems, Discrete Mathematics & Probability

## EXPERIENCE

**Industrial Technology Research Institute** — 11/2025 – 07/2026

*Research Engineer – Generative AI and Computer Vision* — *Hsinchu, Taiwan*

- Developed an **agentic AI pipeline** for automated conformal cooling-channel design, integrating multi-view CAD preprocessing, Qwen3-VL image-to-CadQuery generation, NVIDIA Nemotron channel generation, geometric reconstruction, and STEP validation; **reduced design time by 96%** versus manual design workflows.
- Fine-tuned **Qwen3-VL-32B with LoRA on NVIDIA H200 GPUs** and curated **330K+ CAD training data**, achieving **97% accuracy** on image-to-CadQuery generation tasks.
- Fine-tuned **NVIDIA Nemotron 3 Nano 30B** with NeMo AutoModel and integrated **RAG** to generate parameterized cooling-channel layouts from CadQuery code; engineered end-to-end data normalization, inference, validation, format-reconstruction, and JSON-to-STEP pipelines across NVIDIA DGX Spark and H200 systems.
- Expanded limited industrial-defect datasets with **NVIDIA Cosmos synthetic data** and trained YOLO-based **Automated Optical Inspection model**, **improving defect-detection accuracy by 24%** under low-data.
- Re-engineered an event-driven PLC-triggered IDS uEye optical inspection system using timestamp-based debounce and real-time diagnostics; reduced high-load frame loss from ∼**10 per 683 frames to zero** and eliminated electrical-bounce ghost triggers from ∼**9 per 1,017 frames to zero**.

**Tunghai University** — 06/2024 – 08/2024

*Undergraduate Researcher – Intelligent System Lab* — *Taichung, Taiwan*

- Developed a **GAN-based synthetic-data pipeline** to generate realistic medical samples under data-scarce conditions.
- Optimized membership functions for an **Adaptive Neuro-Fuzzy Inference System(ANFIS)** and applied the model to disease prediction for conditions with unknown etiologies under the supervision of Professor Kuo-Ping Lin.

**Knowledge and Service Information Corp** — 12/2023 – 01/2024

*Software Engineering Intern* — *Taichung, Taiwan*

- Built a database interface that enabled non-technical users to upload and retrieve records without SQL, **reducing reliance on technical staff by 30%**.
- Automated conversion of database records into standardized PDF tables using Java, JavaScript, and SQL, eliminating approximately **one week of manual formatting effort**.

**World Champion, 4th Kibo Robot Programming Challenge** — 03/2023 – 10/2023

*Team Flying Unicorns* — *Japan/Taiwan*

- Programmed **autonomous free-flying robots** for the International Space Station to perform astronaut assistance and maintenance tasks in JAXA's Kibo module.
- Won the **world championship** among **2,700+ contestants from 35+ countries**.

**IBSC Robotics Workshop** — 08/2021 – 08/2023

*Founder & President* — *Taichung, Taiwan*

- Founded and led a robotics program for students in grades 1–12, covering programming and robotics hardware design; coached **3 teams to Distinction awards** at the AERC Asia Robotics Competition.

## PUBLICATIONS

**"Customer Responses for Menu-Less Restaurants under Information Asymmetry"** — Published 06/2022

- Published in *Mathematical Problems in Engineering*, Volume 2022, Article ID 8059340.

**"Solution for Open Questions in Yen (2021) Osler (2001) and Çalışkan (2020, 2022)"** — Published 03/2023

- Published in *IAENG International Journal of Applied Mathematics*, Volume 53(1), pp. 48–57.

**"Optimizing LINE Virtual Assistant's Responses Using Similarity Measure"** — Awarded 2023

- Received **Distinction** in the 2023 Taiwan National High School Project Contest.

## SKILLS

**Languages**: Python, C, C++, Java, SQL, JavaScript, TypeScript, Swift, HTML/CSS, RISC-V

**Technologies**: PyTorch, TensorFlow, Hugging Face, NeMo AutoModel, Qwen3-VL, NVIDIA Nemotron, RAG, YOLO, NVIDIA Cosmos, Pandas, Docker, Git/GitHub, NVIDIA H200 GPU systems, DGX Spark, AWS, React, Expo, SolidWorks, CadQuery, IDS uEye, PLC systems